

Surface Reconstruction from Noisy Point Clouds

Madiyar Bolatov, AAI-2501M

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1 Objective

The objective of this project is to investigate the effect of different regularization terms on the reconstruction of a smooth surface (mesh) from a noisy point cloud. The reconstruction is performed using a deformable mesh model initialized as a spherical surface. The study focuses on comparing different loss function configurations and analyzing their influence on the resulting geometry.

2 Data Preparation

2.1 Object Selection

Instead of the default object from the tutorial, a custom 3D model (sofa-like object) was used. The model was loaded from an .obj file.

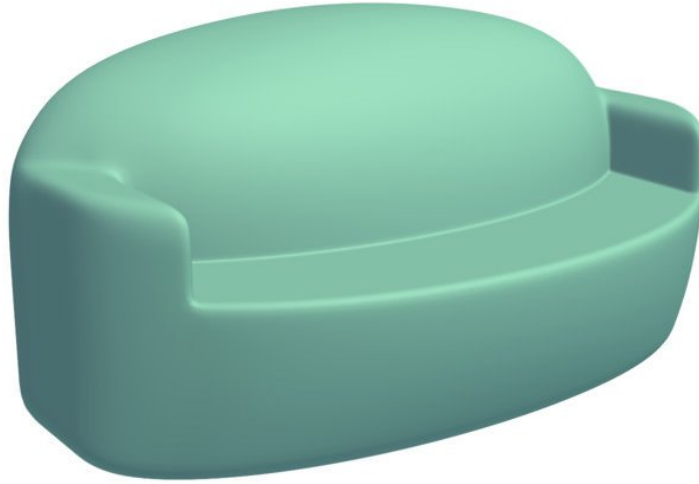


Figure 1: Initial object

2.2 Point Cloud Generation

A point cloud of 5,000 points was sampled from the surface of the mesh using PyTorch3D.



Figure 2: Point Clouds

2.3 Noise Addition

Gaussian noise was added to simulate real-world scanning conditions:

- Noise type: Gaussian
- Standard deviation: 0.01

2.4 Normalization

The point cloud was normalized to: **zero mean (centered)**, **unit scale** This ensures stable optimization.

3 Methodology

3.1 Initial Mesh

A spherical mesh was used as the initial surface:

```
src_mesh = ico_sphere(5, device)
```

The parameter 5 defines the subdivision level and controls the resolution of the mesh.

3.2 Deformation Process

The mesh vertices were iteratively updated using gradient-based optimization:

- deformation variable: vertex offsets
- optimizer: SGD
- iterations: 2000

3.3 Loss function

The total loss is defined as:

$$L = w_c \cdot L_{\text{chamfer}} + w_e \cdot L_{\text{edge}} + w_l \cdot L_{\text{laplacian}}$$

4 Experiments

4.1 Experiment A – No Regularization

$$(w_{\text{chamfer}}, w_{\text{edge}}, w_{\text{laplacian}}) = (1.0, 0.0, 0.0)$$

4.2 Experiment B – Strong Smoothness

$$(w_{\text{chamfer}}, w_{\text{edge}}, w_{\text{laplacian}}) = (1.0, 0.0, 10.0)$$

4.3 Experiment C – Balanced Regularization

$$(w_{\text{chamfer}}, w_{\text{edge}}, w_{\text{laplacian}}) = (1.0, 0.5, 0.05)$$

5 Results and Analysis

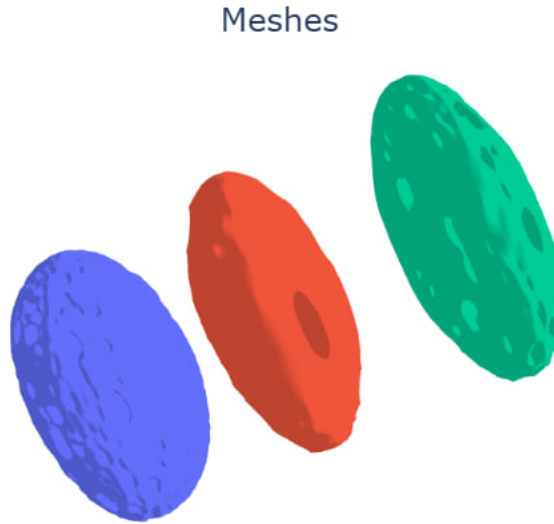


Figure 3: Results of A-blue, B-orange, C-green experiments

1. Experiment A achieves the lowest loss because it optimizes only Chamfer distance. However, it produces **noisy and unstable surfaces**.
2. Experiment B produces the smoothest mesh but fails to preserve important geometric features.
3. Experiment C provides the best compromise between fidelity and smoothness.

6 Conclusion

In this project, a deformable mesh approach for surface reconstruction from noisy point clouds was implemented using PyTorch3D.

The study demonstrates that:

- Chamfer distance alone leads to overfitting and noisy surfaces.
- Strong smoothing improves stability but removes important details.
- A balanced combination of loss terms produces the most visually plausible result.

Additionally, the choice of initial mesh and its resolution plays a crucial role in reconstruction quality. The use of a spherical mesh limits the ability to reconstruct objects with complex geometry.